

Auteur: Niels Hertoghs
Studierichting: Informatica
Oefeningenreeks 7A nr. 12.3

$$f(x, y) = x^3 - x^2y + xy^2 - y^3 \text{ in } a = (1, -1) \text{ in de richting } h(2, 3)$$

$$Df(a, h) = \lim_{\lambda \rightarrow 0} \frac{f((1, -1) + \lambda(2, 3)) - f(1, -1)}{\lambda}$$

$$= \lim_{\lambda \rightarrow 0} \frac{f(2\lambda + 1, 3\lambda - 1) - f(1, -1)}{\lambda}$$

$$f(1, -1) = 4$$

$$f(2\lambda + 1, 3\lambda - 1) = (2\lambda + 1)^3 - (2\lambda + 1)^2(3\lambda - 1) + (3\lambda - 1)^2(2\lambda + 1) - (3\lambda - 1)^3$$

$$= 13\lambda^3 + 28\lambda^2 - 6\lambda + 4$$

$$\Rightarrow \lim_{\lambda \rightarrow 0} \frac{-13\lambda^3 + 28\lambda^2 - 6\lambda + 4 - 4}{\lambda} = \lim_{\lambda \rightarrow 0} (-13\lambda^2 + 28\lambda - 6) = -6$$

References